Multivariate Random Variables

Joint Distributions, Independence, Covariance, and the Multivariate Gaussian

Firuz Kamalov

MAI 600: Mathematics and Statistics for AI

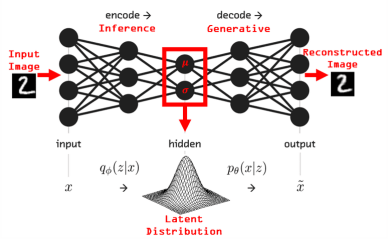
October 24, 2025

Lecture Outline

Why Study Probability Distributions for AI?

1. Generative AI

How does a model generate an image that doesn't exist?



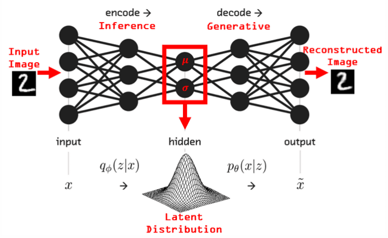
Variational Auto Encoder.

It learns a **probability distribution** from a vast dataset of real images. To create a new image it simply draws a random sample from this distribution.

Why Study Probability Distributions for AI?

1. Generative AI

How does a model generate an image that doesn't exist?



Variational Auto Encoder.

It learns a **probability distribution** from a vast dataset of real images. To create a new image it simply draws a random sample from this distribution.

2. Language Models (LLMs)

How does a model like GPT or Gemini decide on the next word in a sentence?

`"The cat sat on the ___."`

It calculates a **probability distribution** over the entire vocabulary:

- P("mat") = 0.85
- P("chair") = 0.10
- P("rocket") = 0.00001

It then samples from this distribution to select the next word.

Moving from One to Many Variables

Motivating Example: Describing a Student

If we want to describe a student mathematically, one piece of information isn't enough. We might collect several:

- X_1: Score on Exam 1 (e.g., 85)
- X_2: Hours spent studying per week (e.g., 10)
- X_3: Number of lectures attended (e.g., 25)

We can group these into a single vector:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 85 \\ 10 \\ 25 \end{bmatrix}$$

This vector is a **multivariate random variable** (or **random vector**). It's a way to handle multiple random quantities at once.

Joint Distributions: The Idea

A joint distribution tells us the probability of multiple variables *taking specific values at the same time*.

Example: Flipping Two Coins

Let X be the outcome of the first coin (0 for Tails, 1 for Heads), and Y be the outcome of the second.

What is the probability of getting a specific pair of outcomes, like (Heads, Tails)?

$$P(X = 1, Y = 0) = ?$$

The joint distribution gives us the answer for all possible pairs.

Joint PMF (Discrete Example)

For our two-coin flip example, assuming fair coins, the **Joint Probability Mass Function** can be shown in a table:

	Y = 0	Y = 1	Total (Marginal of X)
X = 0	*0.25*	*0.25*	0.5
X = 1	*0.25*	*0.25*	0.5
Total (Marginal of Y)	0.5	0.5	1.0

For example, $P(X = 1, Y = 0) = 0.25$. This table describes the entire probability landscape for the two variables together.

Marginal Distributions

Idea: What if we only care about one variable?

Using our coin-flip table, what's the probability that the first coin is Heads ($X = 1$), regardless of the second coin's outcome?

We just sum across the row for $X = 1$:

$$P(X = 1) = P(X = 1, Y = 0) + P(X = 1, Y = 1) = 0.25 + 0.25 = 0.5$$

This is the **marginal probability**.

Joint PDF (Continuous Example)

What about variables that can take any value, like height and weight?

Example: Height and Weight

Let X be height and Y be weight. We can't create a table for every possible height-weight combination.

Instead, we use a **Joint Probability Density Function (PDF)**, $f(x, y)$, which is a surface. The probability of being in a certain range of height and weight is the *volume under this surface*.

Joint PDF (Continuous Example)

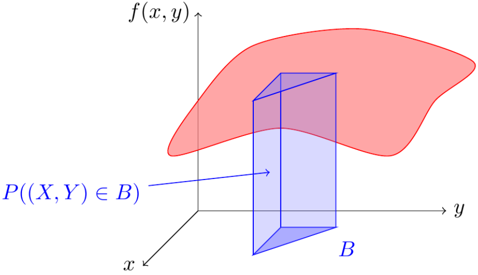


Figure: A joint PDF for two continuous variables.

Calculating Probability with a Joint PDF

Example: Continuous PDF

$$f(x,y) = \begin{cases} x + y & \text{if } 0 \le x \le 1, \text{ and } 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Let's calculate $P(X < 0.5, Y < 0.5)$.

Calculating Probability with a Joint PDF

Example: Continuous PDF

$$f(x,y) = \begin{cases} x+y & \text{if } 0 \le x \le 1, \text{ and } 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Let's calculate $P(X < 0.5, Y < 0.5)$.

$$\begin{aligned} P(0 \le X \le 0.5, 0 \le Y \le 0.5) &= \int_0^{0.5} \int_0^{0.5} (x+y)\,dx\,dy \\ &= \int_0^{0.5} \left[\frac{x^2}{2} + yx\right]_{x=0}^{x=0.5} dy \\ &= \int_0^{0.5} (0.125 + 0.5y)\,dy \\ &= \left[0.125y + 0.25y^2\right]_{y=0}^{y=0.5} = 0.125 \end{aligned}$$

Independence: The Idea

Two random variables are **independent** if the outcome of one tells you nothing about the outcome of the other.

Example: Independent

- X: Result of rolling a red die.
- Y: Result of rolling a blue die.

Knowing the red die is a 6 gives you no information about what the blue die will be. They are independent.

Example: Dependent

- X: A person's height.
- Y: A person's weight.

Knowing someone is very tall makes it more likely they are also heavy. They are dependent.

The Mathematical Rule for Independence

Two random variables X and Y are independent if and only if their joint distribution is the product of their marginal distributions.

For our two-coin flip example:

Is $P(X = x, Y = y) = P(X = x)P(Y = y)$ for all pairs (x, y)?
Let's check for $(X = 1, Y = 0)$:

- From the joint table, $P(X = 1, Y = 0) = 0.25$.
- From the marginals, $P(X = 1) = 0.5$ and $P(Y = 0) = 0.5$.
- $P(X = 1)P(Y = 0) = 0.5 \times 0.5 = 0.25$.

It works! (And it works for all pairs). So, the two coin flips are independent.

Measuring Dependence

We know height and weight are dependent. But *how* are they related?

- As height increases, weight *tends to increase*. This is a **positive relationship**.
- As hours of TV watched per week increases, exam scores *tend to decrease*. This is a **negative relationship**.

Covariance is a measure that quantifies this relationship.

Covariance: The Measure of Joint Variability

The covariance between two random variables X and Y is:

$$\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Interpretation:

- Cov(X, Y) > 0: X and Y tend to move in the **same** direction.
- Cov(X, Y) < 0: X and Y tend to move in **opposite** directions.
- Cov(X, Y) $= 0$: No linear relationship.

Important: If X and Y are independent, their covariance is 0. But if covariance is 0, they are not necessarily independent.

The Problem with Covariance: Scale

Example: Height vs. Weight

- If we measure height in meters and weight in kilograms, we might find Cov(H, W) = 12.
- If we switch to measuring height in centimeters, the value of height is 100 times larger. The new covariance will be much larger too: Cov(H_cm, W) = 1200.

The relationship is the same, but the covariance number changed dramatically! This makes it hard to judge the "strength" of a relationship.

Correlation: A Standardized Measure

The **correlation coefficient (ρ)** solves this problem by normalizing the covariance.

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

This value is always between **-1 and 1**, regardless of the units of X and Y.

- $\rho \approx 1$: Strong positive linear relationship.
- $\rho \approx -1$: Strong negative linear relationship.
- $\rho \approx 0$: Weak or no linear relationship.

Covariance Matrix: Organizing Pairwise Relationships

When we have many variables (like our student example with scores, study hours, and attendance), we can organize all the pairwise covariances into a **covariance matrix (Σ)**.
For $\mathbf{X} = [X_1, X_2, X_3]^T$:

$$\Sigma = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \mathrm{Cov}(X_1, X_3) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \mathrm{Cov}(X_2, X_3) \\ \mathrm{Cov}(X_3, X_1) & \mathrm{Cov}(X_3, X_2) & \mathrm{Var}(X_3) \end{bmatrix}$$

- The **diagonal** contains the variance of each individual variable ($\mathrm{Var}(X) = \mathrm{Cov}(X, X)$).
- The matrix is **symmetric** since $\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$.

The Multivariate Gaussian: A Bell Curve in Higher Dimensions

- Many natural phenomena (like human height, measurement errors) follow a bell-shaped curve, called the **Gaussian** or **Normal** distribution.
- When we have multiple variables that are jointly distributed in a similar "bell" shape, we have a **Multivariate Gaussian Distribution**.
- It is perhaps the most important distribution in all of statistics and machine learning.

Defining a Multivariate Gaussian

A multivariate Gaussian distribution is completely described by just two parameters:

1. **The Mean Vector (μ):** An n-dimensional vector that tells us the "center" or peak of the bell shape.

$$\mu = E[\mathbf{X}] = \begin{bmatrix} E[X_1] \\ \vdots \\ E[X_n] \end{bmatrix}$$

2. **The Covariance Matrix (Σ):** An $n \times n$ matrix that tells us the "spread" and "orientation" of the bell.

Visualizing the Covariance Matrix Effect

For a 2D Gaussian, the covariance matrix Σ controls the shape of the density's contours.

Case 1: No Correlation

If $\mathrm{Cov}(X_1, X_2) = 0$, the variables are uncorrelated. The contours of the Gaussian are circles or axes-aligned ellipses.

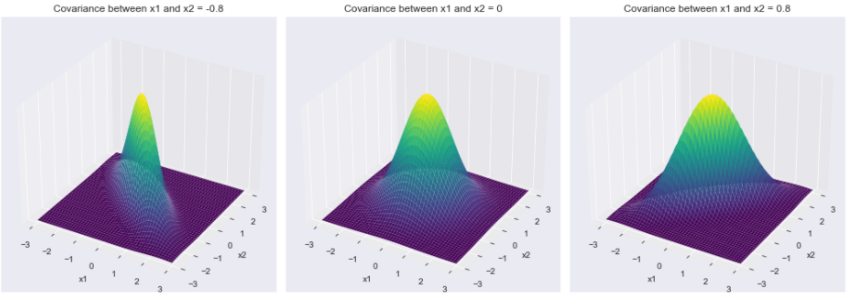
Case 2: Positive Correlation

If $\mathrm{Cov}(X_1, X_2) > 0$, the variables tend to increase together. The contours are ellipses tilted in a positive-slope direction.

Case 3: Negative Correlation

If $\mathrm{Cov}(X_1, X_2) < 0$, the variables tend to move in opposite directions. The contours are ellipses tilted in a negative-slope direction.

Visualizing the Covariance Matrix Effect



The PDF of the Multivariate Gaussian

Now that we have the intuition, here is the formula for the PDF:

$$f(\mathbf{x}; \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)\right)$$

- You can see the mean μ and covariance Σ are the key parameters.
- The term $(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)$ is called the *Mahalanobis distance*—it measures the distance of a point $\mathbf{x}$ from the mean μ, accounting for the covariance structure.

AI Application 1: Anomaly Detection

Scenario

We are monitoring two metrics from a server: CPU usage (X_1) and Memory usage (X_2). Under normal operation, these metrics follow a 2D Gaussian distribution.

AI Application 1: Anomaly Detection (cont.)

- We model the "normal" behavior by fitting a multivariate Gaussian to a dataset of normal operational data.
- A new data point (a new measurement of CPU/Memory) arrives.
- We can calculate its probability density under our model.
- **If the probability is extremely low**, the point is far from the center of the "normal" cluster. We can flag it as a potential **anomaly** (e.g., a system failure or cyberattack).

AI Application 2: Generative Models

Scenario

Imagine we have a large dataset of images of handwritten digits (e.g., the number "8"). Each image is $28 \times 28 = 784$ pixels. We can treat each image as a point in 784-dimensional space.

- We can model the probability distribution of these 784-pixel vectors using a multivariate Gaussian. We compute the mean image (μ) and the 784×784 covariance matrix (Σ).
- **To generate a new image of an "8"**, we simply draw a random sample from this learned Gaussian distribution $N(\mu, \Sigma)$.
- The result is a new 784-dimensional vector that, when reshaped into an image, will look like a plausible, novel handwritten "8".

AI Application 3: Neural Network Weight Initialization

The Problem: How to Start Training a Neural Network?

- **Initializing to zero:** Causes all neurons in a layer to learn the exact same thing.
- **Initializing with large values:** Can cause gradients to explode or vanish, stopping the learning process.

AI Application 3: Neural Network Weight Initialization

The Problem: How to Start Training a Neural Network?

- **Initializing to zero:** Causes all neurons in a layer to learn the exact same thing.
- **Initializing with large values:** Can cause gradients to explode or vanish, stopping the learning process.

The Solution: Use a Scaled Normal Distribution

The standard practice is to initialize weights by drawing random values from a Gaussian distribution with a mean of 0.

$$W \sim \mathcal{N}(0, \sigma^2)$$

The key is to choose the variance carefully to keep the signal flowing smoothly through the network.

AI Application 3: Neural Network Weight Initialization

State-of-the-Art Initialization Methods

Modern techniques automatically scale the variance based on the layer's size:

- **Xavier/Glorot Initialization:** Sets the variance $\sigma^2 = \frac{2}{n_{in}+n_{out}}$. Works well for activations like 'tanh'.
- **He Initialization:** Sets the variance $\sigma^2 = \frac{2}{n_{in}}$. This is the standard for the very popular ReLU activation function.

These methods are direct applications of the normal distribution and are critical for training deep neural networks.

Lecture Summary

- **Multivariate Random Variables:** Allow us to model multiple related quantities in a single vector.
- **Joint Distributions:** Describe the probability of multiple variables occurring together.
- **Independence:** A powerful simplifying assumption where variables have no influence on each other.
- **Covariance and Correlation:** Measures the strength and direction of the linear relationship between two variables.
- **Multivariate Gaussian Distribution:** A high-dimensional bell curve defined by a mean vector and a covariance matrix. It's a workhorse for modeling real-world data in AI.

Thank you. Questions?